

The Lancet - Article Draft (Fictitious)

Clinical features and outcomes of the first 70 patients hospitalised with Spooky-like Virus 2024 (SPOK-24) infection in Brisbane, Australia: a prospective observational study

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Summary

Background A recent cluster of respiratory illnesses in Brisbane, Australia, has been attributed to a novel paramyxovirus, provisionally named Spooky-like Virus 2024 (SPOK-24). We describe the clinical characteristics, outcomes, and complications among the first hospitalised patients with laboratory-confirmed SPOK-24 infection.

Methods In this prospective observational study, we enrolled the first 70 patients with RT-PCR-confirmed SPOK-24 infection who were admitted to hospitals in the Brisbane Metro area between Sept 26 and Oct 20, 2024. Data were collected using a modified WHO/ISARIC clinical characterization protocol. Variables included demographic data, symptom onset, clinical findings, laboratory and radiologic results, treatments, and outcomes. Outcomes were compared between patients who required intensive care unit (ICU) admission and those who did not.

Findings Of 70 patients, 66 (94%) presented with fever, 48 (69%) with non-productive cough, and 31 (44%) with myalgia or fatigue. Dyspnoea developed in 23 (33%) of patients, with a median time from symptom onset to dyspnoea of 6.0 days (IQR 4.0–9.5). Other symptoms included sore throat (26%), rhinorrhoea (20%), and headache (14%). Neurological symptoms were noted in four patients (6%), including two with confirmed encephalitis.

Abnormal chest imaging was found in 67 patients, with bilateral infiltrates in 87% of them. Laboratory findings showed lymphopenia in 49% and elevated C-reactive protein in 81%. Thirteen (19%) patients required ICU admission, five (7%) required mechanical ventilation, and six (9%) died. Complications included acute respiratory distress syndrome (9%), encephalitis (3%), and secondary bacterial pneumonia (6%).

Interpretation SPOK-24 is associated with a spectrum of clinical disease, from mild flu-like illness to severe pneumonia and neurologic complications. Although case fatality was low in this cohort, the occurrence of encephalitis and acute respiratory failure underscores the need for continued vigilance. Further studies are

required to characterise long-term outcomes, optimal management strategies, and the impact of early interventions.

Funding Australian Department of Health; Global Zoonosis Preparedness Consortium.

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